GPE Week 5:

Exchange Rates and
The International Movement of Capital

Introduction to Exchange Rates

- This week, we will learn all about exchange rates, and why they matter
- This is a phenomenon that of you will encounter, at least a little bit, when you enter or leave the Eurozone and have to change your money from one currency to another
- For international business and finance, however, exchange rates are of fundamental importance
- Generally speaking, businesses prefer stable exchange rates
- This is because they must sign contracts, which might take months or years to gain receipts for

Introduction to Exchange Rates

- If during that time exchange rates fluctuate, you could either double your profit (which will annoy your customer), or you might lose all your profit (which will mean you did the deal for nothing)
- Since companies often operate at a low profit margin (of a few percent), even a small movement in exchange rates can wreak havoc with their bottom line
- For this reason, companies **hedge** by trying to lock in **forward rates** which both parties can agree upon
- (Hedging is an attempt to reduce **exchange rate exposure**—which means, you are exposed to risk of a changing exchange rates)
- But all of this costs time and effort, and involves risk. It therefore costs money, and reduces the incentives to do business at all

Introduction to Exchange Rates

- As you can see, then, MNEs must be concerned with the movement of exchange rates
- Exchange rates are also intimately connected to trade deficits, since as we saw last week, foreign capital is required to fund trade deficits
- And interest rates also repel or attract international capital
- Also, financiers must take all of these things into account when moving large sums, for the same reasons as MNEs
- Finally, Investors can try to take advantage of movements in exchange rates
- The whole idea of investing in foreign currencies (Forex markets) is based around the idea that you buy when a currency is cheap, and then sell it when it is more expensive—and your *expectations* of what will happen in the future
- This is easily the biggest international investment market, which now dwarfs the regular stock market

Introduction to Exchange Rates

- This week, we look at three chapters, which introduce different aspects of exchange rates
- Chapter 16 introduces the concept of exchange rates, and related basic ideas, to familiarize you with the world of foreign exchange
- Chapter 17 then talks specifically about movements of capital which occur when a country has “flexible” exchange rates
- This is the regime that most of the world’s major currencies use
- Chapter 18 finishes by looking at “fixed” exchange rates
- And gives some rationale for why countries might wish to fix their exchange rates, and warnings about the dangers when it does

What is an exchange rate?

- As you might have noticed when travelling internationally, the value of currencies rise and fall against each other
- You will have heard things like “the Yen is strong,” “the Bhat is weak,” or “the Euro is rising against the pound.”
- Well, what does this mean?
- The main definition of an exchange rate is the **price of a currency**.
- You can also call this the **value of a currency**. A currency unit that can be bought for a Eurocent is worth less than one which takes a Euro or more to purchase. But what matters of course, is the trend, rather than the absolute value. Is a currency getting more valuable, or less valuable?
- What also matters is the **total amount** of a currency in circulation. If there are one billion Zimbabwean dollars in circulation, but all of these can be purchased for one US dollar, than the total value of the ZWD is 1 USD. Each individual ZWD is worth only 1/1,000,000,000 USD.

What is an exchange rate?

- Usually, investors will price a currency the same, even if more are printed. E.g. if Zimbabwe printed 2 billion total ZWD, they would likely stay being worth 1USD total, meaning the value of each ZWD would slip to 1/2,000,000,000 of a USD
- This is because if you print more money, but the country's GDP output (Y) stays the same, then we say that "more money is chasing the same amount of goods" which means that the price of everything goes up internally;
- Meanwhile, the value of individual currency units will go down, i.e. you will have to pay more of them to buy a foreign currency unit, and/or foreigners will have to pay less to buy your units, when capital is moved to and from that country across currency units

What is an exchange rate?

- We therefore see the importance of the quantity of a currency, in determining its overall value
- We see how printing (creating) money very quickly causes inflation.
- While, more slowly, destroying currency can lower inflation.
- Note also that for a currency to have a meaningful price, it has to be priced in terms of another accepted currency—it's logically absurd to quote the price of Yen in Yen, or USD in USD.
- For many decades, currencies were priced in terms of gold, because gold was rare and produced at a slow, steady rate.
- Since WWII, most global currencies have been priced in terms of the currency of the global hegemon, i.e. the USD.
- Of course, you can price a global currency in terms of any other currency, or sometimes according to a "basket of currencies."

Why are exchange rates important?

- We have hinted last week that exchange rates are essential to global trade and finance
- Exchange rates help determine whether an economy is attractive to international investors
- If investors believe that a currency is strong and likely to fall, then they will be less likely to invest
- If they believe a currency is weak and likely to increase in value, they are more likely to invest
- For example, if this week the USD = 1 EUR, but you know that next week the USD will be worth 2 EUR, then we say that the USD is now weak and it's going to get stronger against the EUR. It would be smart to spend 1000 EUR on 1000 USD now, so that next week, you could exchange that 1000 USD for 2000 EUR. In one week you will double the number of EUR in your bank account.
- If the currency is likely to do the opposite, then next week your investment will only be worth half as many EUR.

Why are exchange rates important?

- But exchange rates are much more important than simply as an investment opportunity
- In general, currencies **appreciate** when foreign capital flows in, and they **depreciate** when foreign capital flows out.
- But they do not only respond to the inflow of foreign capital, as we've said, they also help determine whether investment is attractive.
- They help determine the amount of FDI and foreign portfolio investment
- They help determine the amount of foreign capital that can pay for imports
- They therefore help determine trade deficits, as we saw last week
- International investors are attracted by high interest rates; therefore, exchange rates are affected by interest rates
- Exchange rates affect interest rates, and also interest rates are affected by exchange rates

Why are exchange rates important?

- Also, exchange rates are affected by **fundamentals**
- Investors tend to stay away when a government is being irresponsible. A big part of the appeal of the USD and the EUR is that the regimes that prop them up are seen as being governed responsibly (*cough*)
- A country with a high budget deficit will usually suffer from a weak currency
- Countries which borrow too much will see their currency weaken
- Countries which do not tax enough likewise
- High inflation naturally scares investors
- High trade deficits will also scare investors and make currencies weak
- Exchange rates are therefore a major economic indicator that policymakers have to take into account when making major budgetary and spending decisions
- Exchange rates also determine the rate of exports, and imports (how?)

Exchange Rates Definitions

- The simplest measure of an exchange rate is the
- **Nominal Exchange Rate (e)**
- e= Home Country Currency Price of one unit of foreign currency
- e= X units of Home Currency/1 Unit of Foreign Currency

- Ok, now think about it: **when e goes up**, this means, **the value of your currency goes down**
- Because if it takes 1 home units to buy 1 foreign unit, and this raises to 10 home units to buy 1 foreign unit (e increases), then the value of your currency has gone down from 1 foreign unit to .10 foreign units—it is worth only 1/10th of what it was before
- And when **e goes down**, the value of your **currency goes up**
- (For this reason, we sometimes use 1/e, because this coordinates with your logical faculties better: as 1/e goes up, then your currency increases in value)

Exchange Rate Fractions…

- If you are in Canada, and you see USD e=1.25
- This means, that 1.25 CAD is required to buy 1 USD
- This means that 1 CAD is worth : 80 US cents (1/1.25)
- In the US, you would say that CAD e=.80 because 80 US cents is required to buy 1 CAD
- Or, you can say that 1 USD is worth: (1/.8) = 1.25 CAD
- But we don’t work with “how much is my currency worth” very much; we mostly work with e or 1/e

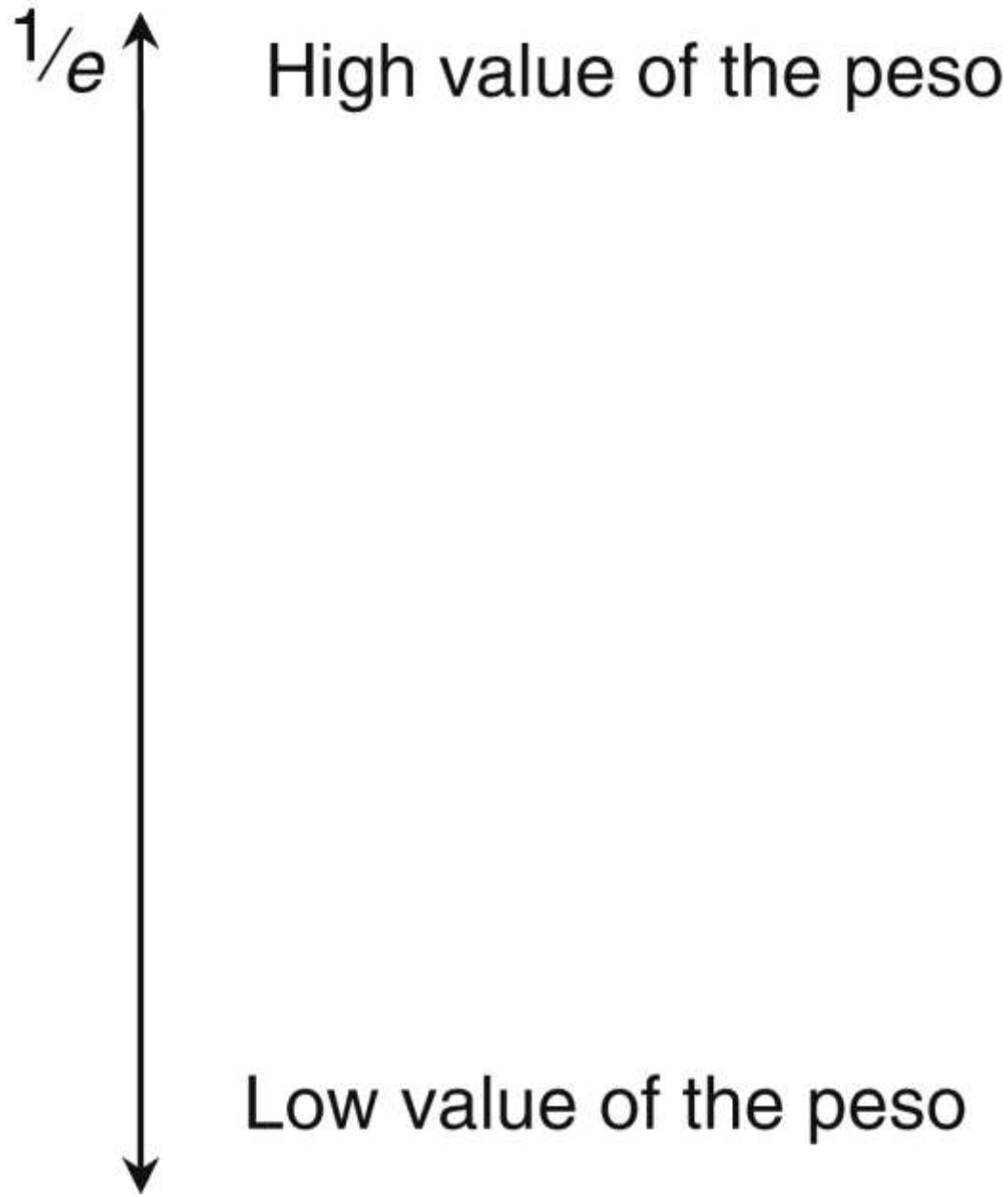
$$e = \frac{pesos}{dollar} \tag{14.1}$$

which is in the form of:

$$e = \frac{home\ currency}{foreign\ currency} \tag{14.2}$$

As implied in these equations, we use the symbol e to denote the nominal exchange rate. Nominal exchange rates for a sample of 15 countries are presented in Table 14.1 for April 2010 and for one year earlier.

$e \uparrow \Rightarrow$ value of the home currency (peso) *falls* or *depreciates*
$e \downarrow \Rightarrow$ value of the home currency (peso) *rises* or *appreciates*


1/e
High value of the peso
Low value of the peso

Country or region	Currency	Nominal exchange rate	Nominal exchange rate 1 year earlier
Argentina	Peso	14.9	8.68
Brazil	Real	3.98	2.83
Canada	Dollar	1.37	1.24
Chile	Peso	701	621
China	Yuan	6.53	6.26
Eurozone	Euro	0.90	0.88
Indonesia	Rupiah	13,510	12,770
Japan	Yen	114	119
Mexico	Peso	18.4	14.9
Pakistan	Rupee	105	102
Russia	Ruble	75.3	63.2
South Africa	Rand	15.5	11.7
Thailand	Baht	35.6	32.6
Turkey	Lira	2.95	2.45
United Kingdom	Pound	0.70	0.65

Yen per US dollar
400
350
300
250
200
150
100
50
0
1960
1962
1964
1966
1968
1970
1972
1974
1976
1978
1980
1982
1984
1986
1988
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018

Exchange Rates

- In Figure 16.2, are we looking at e, or 1/e?
- When the graph goes down, this means that the yen is…
- When the graph goes up, then the yen is…
- What does it mean that 360 yen buy 1 dollar?
- What does it mean when 100 yen buy one dollar?
- What is better for the Yen, e=360 or e=100?
- How much is a yen worth? 1/360 vs. 1/100
- .0027 = .003, vs. .01
- 1/3 cents vs 1 cent
- So the Yen appreciated, … this graph takes the form X Yen/1 USD meaning that it is e from the POV of Japan. As e (the number of home units) goes down against 1 foreign unit, the value of the currency goes up.

Effective Exchange Rate

- This is the nominal exchange rate
- Weighed against a basket of foreign currencies
- Where the weights are determined by the amount of foreign trade that the home currency has with foreign currencies
- So this is a way of saying what the average “real” (vs. the nominal) exchange rate is, given the value of this currency relative to the other currencies that it is normally exchanged in
- This is a way of “triangulating” the value of a currency, against a number of other currencies – this gives a more overall sense of its worth at the moment

This leads us to consider **the effective exchange rate.**[6] To understand this, let's turn back to Mexico and its relationships to the United States and the European Union. Mexico's effective exchange rate (e^{eff}) would be:

$$e^{eff} = a_{US}e_{dollar} + a_{EU}e_{euro} \tag{14.3}$$

Here, e_{dollar} and e_{euro} are the $\frac{peso}{dollar}$ and the $\frac{peso}{euro}$ nominal exchange rates, respectively. The terms a_{US} and a_{EU} are weights that sum to one and determine how much the two nominal exchange rates affect e^{eff} as they change in value. The standard means of determining the weights in Equation 14.3 is via bilateral trade volumes typically measured as the sum of imports and exports to the United States and the European Union, respectively, as a proportion of Mexico's total imports and exports. Recognizing the increasing role of financial relationships in the world economy, however, there has been increased consideration given to asset-related determination of the weights. But the key thing for you to understand is that the effective exchange rate is a means of generalizing the two-country nominal exchange rate for Mexico or any other country across a number of relevant partner countries.

Real Exchange Rate

- The Real exchange rate (**re**), factors in the **price of goods** in each country
- The Real exchange rate, tries to measure the purchasing power of your currency—to anchor it against real things you can exchange it for
- The price of goods, remember, is determined by the inflation rate
- If a country experiences hyperinflation, then you would expect its exchange rates to adjust
- I.e., if it now costs 10 EUR to buy bread, instead of 1 EUR, this means the value of the EUR has fallen 10x
- e will often follow inflation trends
- But before e adjusts, there can be a change in the 'real' exchange rate as opposed to nominal
- re can be measured (but does not have to be measured) as the amount of goods it takes to purchase goods in a foreign currency: i.e., how many US coffeemakers does it take to buy a Mexican bulldozer; normally the rule is, the goods ratio (and thus the real exchange rate) will tend to stay similar, even if the nominal exchange rate fluctuates wildly

Change	Intuition	Effect in "*re*" equation
P^{US} increases	US goods increase in price. Therefore, it takes more Mexican goods to buy a unit of US goods. The real value of the peso has fallen.	Since it is in the numerator, the increase in P^{US} increases the value of *re*.
P^{M} increases	Mexican goods increase in price. Therefore, it takes fewer Mexican goods to buy a unit of US goods. The real value of the peso has risen.	Since it is in the denominator, the increase in P^{M} decreases the value of *re*.
e increases	It takes more Mexican pesos to buy US dollars. The real value of the peso has fallen.	The increase in *e* increases the value of *re*.

Closely related to the definition of the real exchange rate is a model of exchange rate determination known as **purchasing power parity** (PPP). The PPP approach to exchange rates begins with the hypothesis that *the nominal exchange rate will adjust so that the purchasing power of a currency will be the same in every country*. This hypothesis is also worth putting in a box:

> **PPP hypothesis:** The nominal exchange rate will adjust so that the purchasing power of a currency will be the same in every country.

Let's explore the implications of this hypothesis. The purchasing power of a currency in a given country is inversely related to the price level in that country. For example, the purchasing power of the peso in Mexico can be expressed as $\frac{1}{P^M}$. The higher the price level in Mexico, the lower the purchasing power of the peso there. The purchasing power of the peso in the United States is a bit more complicated. First, we need the rate at which a peso can be exchanged into dollars. This is $\frac{1}{e}$. Second, we need the purchasing power of a dollar in the United States. This is $\frac{1}{P^{US}}$. Putting these together, we have the purchasing power of a peso in the United States as $(\frac{1}{e}) \times (\frac{1}{P^{US}})$. Now, we can state the PPP hypothesis as follows:

$$\frac{1}{P^M} = \frac{1}{e} \times \frac{1}{P^{US}} \tag{14.7}$$

The maths…

- Remember, that when a numerator increases, that value increases
- 2/2 is bigger than 1/2
- When a denominator increases, the value decreases
- ½ is bigger than ¼
- So when they say that the value of a currency is inversely related to its price level
- An inverse relationship in maths can be measured as 1/X
- This means that the purchasing power of a currency can be seen as 1/P
- Where the higher the price level, then the less the purchasing power

PPP Model

- The PPP model argues that currency values will accommodate changes in price levels
- It therefore assumes that the real exchange rate does not change
- And yet in reality it does change (albeit not too wildly)
- The PPP model assumes all goods are internationally traded
- But, many goods are not internationally traded
- Nontrade goods
- Locally supplied services, cleaning, repairs, food prep, haircuts
- This weakens the PPP relationship
- But there are other factors, such as closed borders

The Big Mac Index

In 1986, *The Economist* began to publish an annual test of the PPP theory of exchange rates based on an unusual measure of price levels around the world. This "Big Mac index" measures price levels using just one good, the McDonald's Big Mac hamburger. *The Economist* (2007) described its efforts as follows: "The Big Mac index is based on the theory of purchasing power parity (PPP), according to which exchange rates should adjust to equalize the price of a basket of goods and services around the world. Our basket is a burger: a McDonald's Big Mac." Calculations of this index are given in the accompanying table.

	Big Mac Prices				
Country	In Local Currency	In Dollars	Implied PPP	Actual 2009 exchange rate	Local currency over(+)/ under (–) valuation (%)
United States	$3.54	3.54	–	–	–
Argentina	Peso 11.50	3.30	3.25	3.49	–7
Brazil	Real 8.02	3.45	2.27	2.32	–2
Britain	£2.29	3.30	1.55[a]	1.44[a]	–7
Canada	C$ 4.16	3.36	1.18	1.24	–5
Chile	Peso 1,550	2.51	438	617	–29
China	Yuan 12.5	1.83	3.53	6.84	–48
Euro Area	€3.42	4.38	1.04[a]	1.28[a]	24
Indonesia	Rupiah 19,800	1.74	5,593	11,380	–51
Japan	¥290	3.23	81.9	89.8	–9
Mexico	Peso 33.0	2.30	9.32	14.4	–35
Philippines	Peso 98.0	2.07	27.7	47.4	–42
Russia	Ruble 62.0	1.73	17.5	35.7	–51
South Africa	Rand 16.95	1.66	4.79	10.2	–53
South Korea	Won 3,300	2.39	932	1,380	–32
Thailand	Baht 62.0	1.77	17.5	35.0	–50
Turkey	Lire 5.15	3.13	1.45	1.64	–12

[a] Dollars per pound and per euro. *Note:* The exchange rates may differ from those of Table 14.1, because the latter are more recent.

Value of Peso & Trade Deficit

- Now we can begin to look at how exchange rates relate to the trade flows (deficits and surpluses) that we discussed last week
- In this model, there is a positive simple correlation between the value of the currency and the size of the trade deficit
- 1/e means that as Y axis increases, value of the peso goes up
- As peso appreciates, Mexican exports become more expensive for foreigners, and decrease in volume
- (Cheap currency = more exports)
- At the same time, imports become cheaper for Mexican consumers, so Mexicans demand more imports
- So as Peso appreciates, Imports increase in volume
- Z – E means, If Imports (Z) are greater than exports (E), Z-E will have a positive value
- So, as the value of the peso goes up, the value of imports relative to exports go up

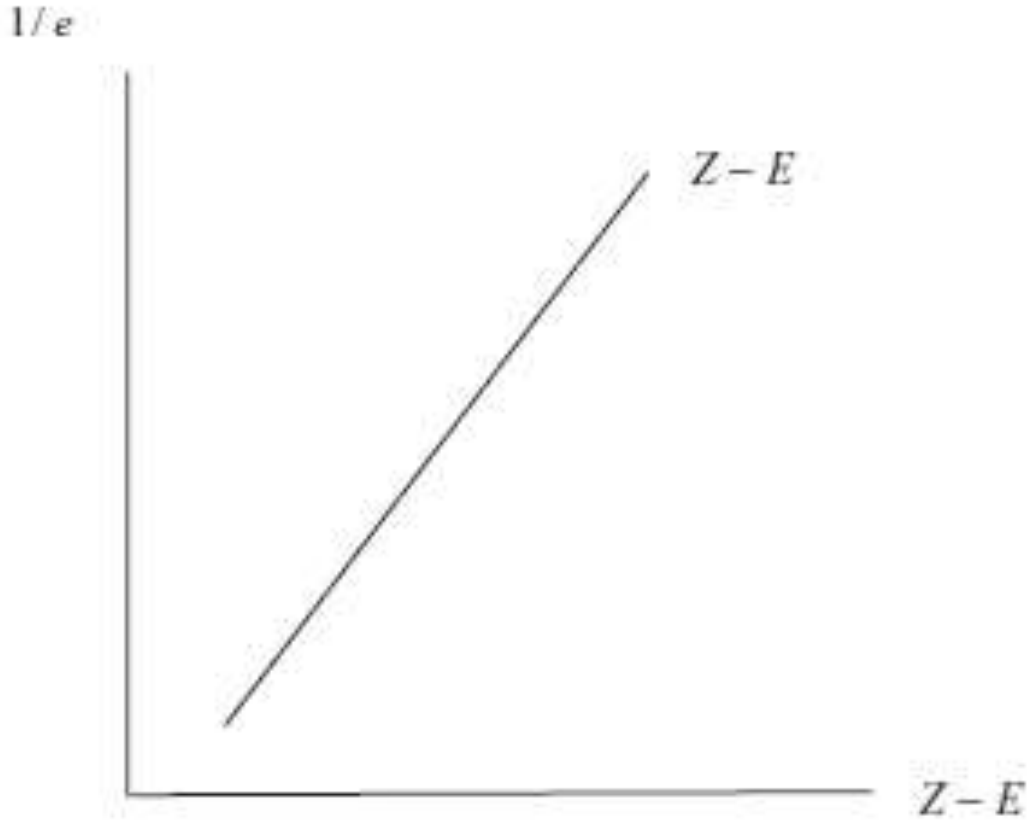


Figure 14.3. The Value of the Peso and Mexico's Trade Deficit

EXCHANGE RATES AND TRADE FLOWS

Changes in e, and therefore in the value of the peso, have an impact on trade flows that is important for you to understand. To see this, we consider the case of Mexico's imports and exports. World prices (P^W) are typically in U.S. dollar terms, and Mexican prices (P^M) are in peso terms. The relationship between the Mexican peso and world dollar prices of Mexico's import (Z) goods can be expressed as:

$$P_Z^M = e \times P_Z^W \tag{14.11}$$

P_Z^W is in dollar terms. Multiplying it by e gives us P_Z^M in peso terms. Now, suppose that e were to increase (the value of the peso falls). This movement down the scale in Figure 14.1 increases the peso price of the imported good in Mexico. Following the "law of demand," import demand consequently decreases.[11] Next, suppose e were to decrease (the value of the peso rises). This movement up the scale in Figure 14.1 decreases the peso price of the imported good in Mexico. Import demand consequently increases.

Next, consider the case of Mexico's exports. The relationship between the peso and dollar prices of Mexico's exported (E) goods can be expressed as:

$$P_E^M = e \times P_E^W \tag{14.12}$$

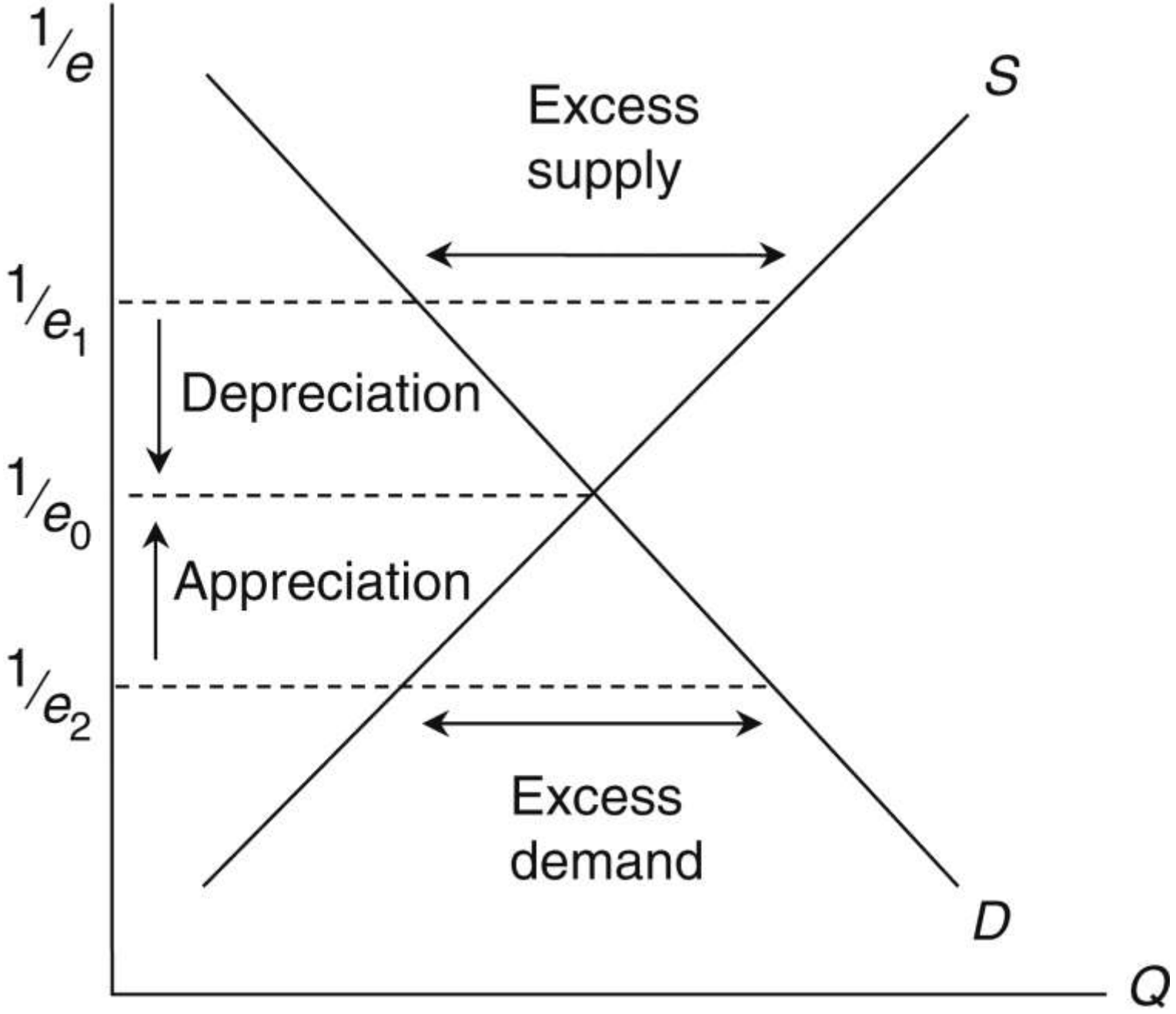
In this equation, P_E^M is the price of Mexican export goods. For a given P_E^M, if e were to increase (the value of the peso falls), P_E^W will fall to maintain equality. The movement down the inverse scale in Figure 14.1 would consequently cause a fall in the world price

of Mexico's exports. This increases foreign demand for Mexico's goods, and Mexico's exports consequently increase.[12] Next, suppose that e were to decrease (the value of the peso rises). This movement up the inverse scale in Figure 14.1 increases the world price of Mexico's exports, and export demand would consequently decrease.[13]

What we have seen here is that, as the nominal exchange rate increases or the value of the peso falls, imports decrease and exports increase. We can represent these relationships as one between the value of the peso and the trade deficit, $Z - E$. This is done in Figure 14.3. As you can see, there is a *positive* relationship between the value of the peso and the trade deficit. We use this relationship in the remaining chapters of the book in this section on international finance.[14]

Marketplaces for Funds

- In Chapter 17 we develop two models which attempt to explain how exchange rates fluctuate in the real world
- They assume a “flexible regime” (i.e., floating currency regime), in which markets are allowed to determine the value of a currency
- In the following graph, we see some basics of how a currency market works
- Money is subject to supply and demand forces, like everything else
- If people demand a currency, it becomes more scarce, and its value increases (and vice versa)
- The following demonstrates a basic market for pesos

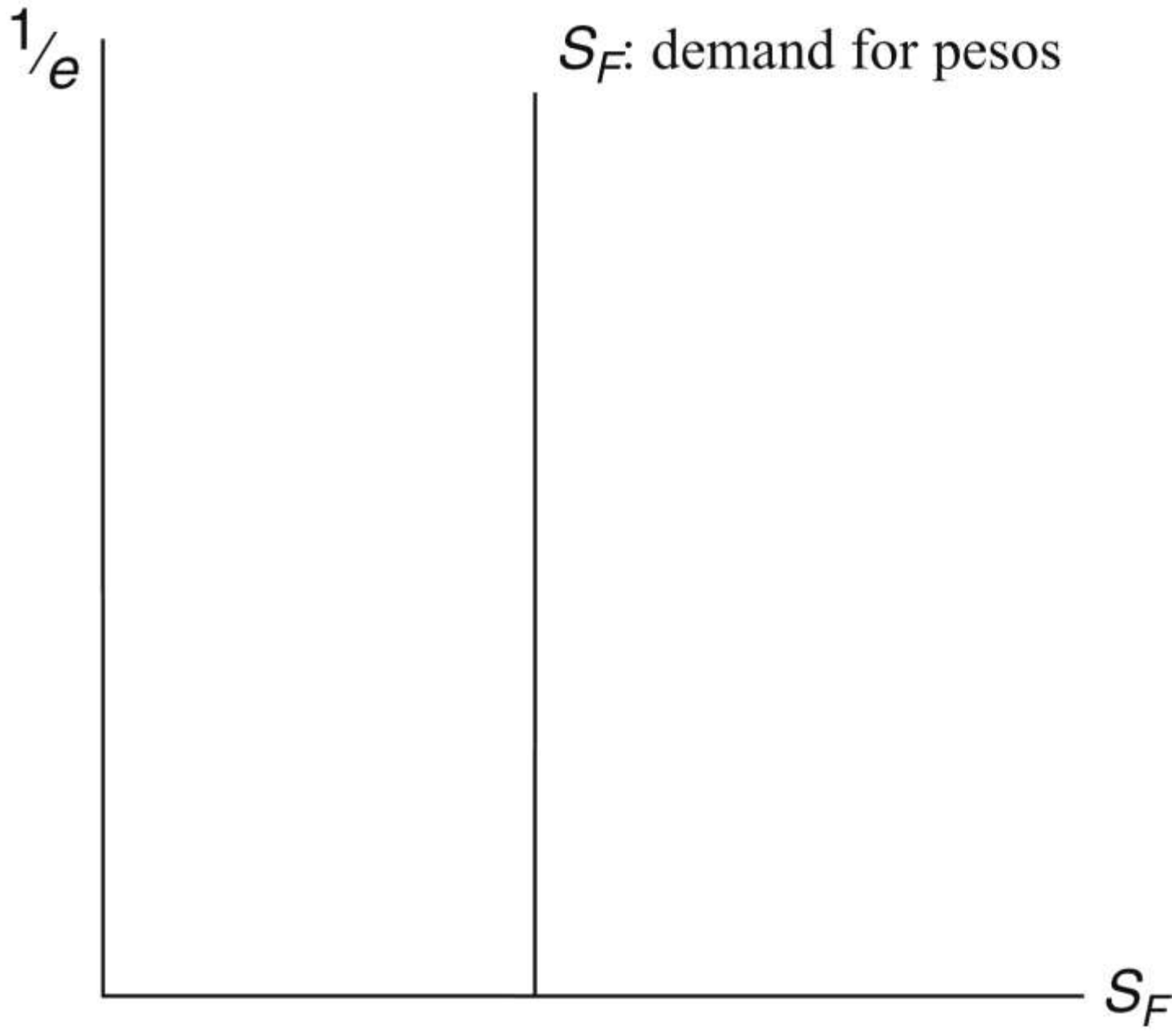

1/e
S
Excess
supply
1/e1
Depreciation
1/e0
Appreciation
1/e2
Excess
demand
D
Q

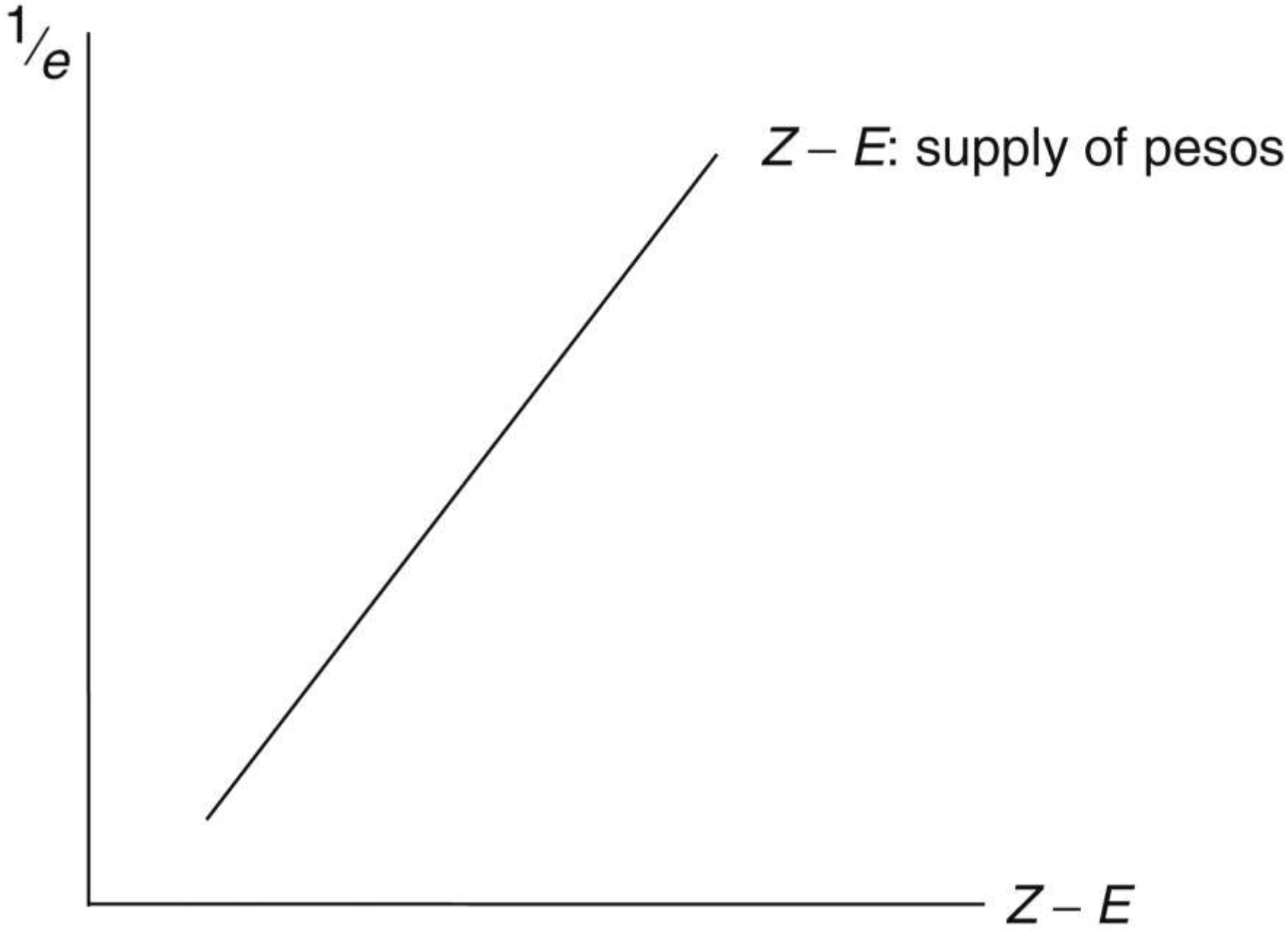
Trade-Based Model

- The question we now ask is: what drives the movement of exchange rates?
- We will first play with a trade-based model, which suggests that changes in trade deficits can affect the movement of exchange rates (and vice versa)
- To illustrate, we remember our accounting equation from last week, which says that Z-E = Sf
- That is to say, that your trade deficit will be financed by (and equal to), foreign savings
- Foreign savings, it turns out = a major source of demand for your currency
- Because foreigners, to spend in your economy, have to turn their money into your money, meaning they increase demand for your money

Trade-Based Model

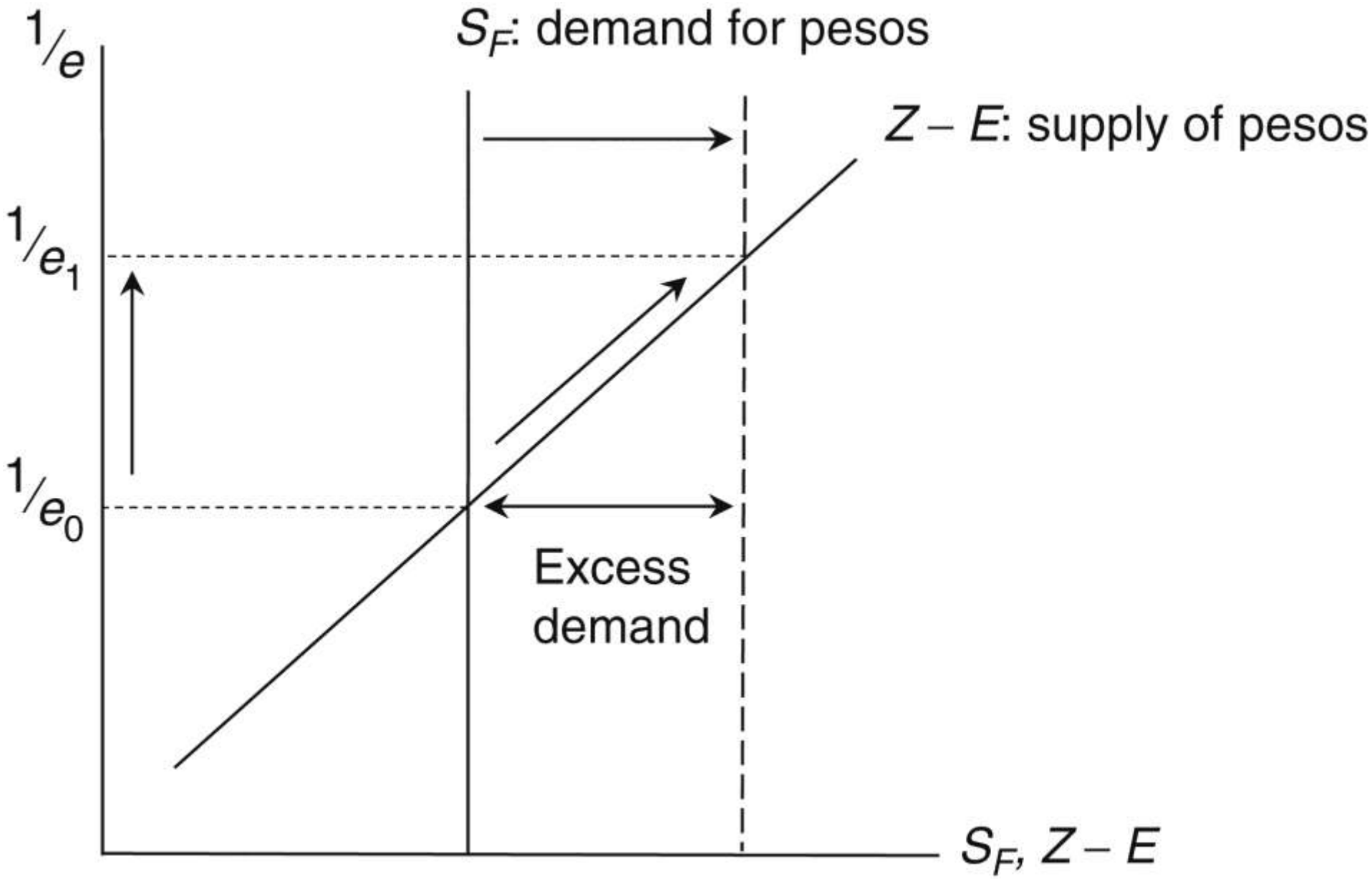
- In our Trade Based model, we assume that foreign demand for your currency is irrespective of its value (it depends on demand for goods, rather than on exchange rates)
- So the demand curve is inelastic, with respect to currency value, as seen in the next slide
- The supply however, is a little different
- We have seen that as the value of currency expands, so does the trade deficit (exports more expensive, imports cheaper)
- As Mexicans buy imports, they demand fewer pesos, because they exchange pesos for dollars
- So, as the trade deficit goes up, the supply of pesos increases


1/e
S_F: demand for pesos
S_F


1/e
Z – E: supply of pesos
Z – E

Trade-Based Model

- Putting S and D together, we get an equilibrium price of 1/e = 0
- Then, imagine that an inflow of foreign savings comes into the country
- Other things equal, this will shift the Sf curve (D) to the right
- This means that the value of the peso will appreciate
- Because the trade deficit will expand as exports are not wanted, leading the value of pesos to increase along the supply curve
- (The supply curve is positively sloped on the idea that when Mexicans import more, they sell pesos and demand dollars)
- So an inflow of foreign capital leads to increases in the value of your currency
- Capital outflows will have the opposite effect
- The model is *trade based*, b/c it assumes that only trade deficits change in relation to a change in the value of the peso


$1/e$
S_F: demand for pesos
$Z - E$: supply of pesos
$1/e_1$
$1/e_0$
Excess demand
S_F, $Z - E$

Assets-Based Model

- Remember our fundamental equation Sf = Z-E
- In the trade-based model, we were focusing on the trade deficit portion of this equation
- Now, we focus on the Sf version: i.e., we look at the movement of funds across borders
- This is mostly done by trading financial “assets” therefore we call this an Assets-Based Model
- For this model, we consider the perspective of a Mexican investor

Assets-Based Model

- For a Mexican investor, investing in peso-denominated assets is straightforward: the return = the interest rate
- But investing in USD-denominated assets means that you have to consider not only the interest rate, but also the possible exchange rate
- If the peso **depreciates** against the dollar, you will have a **capital gain** in terms of pesos
- So, you invest in foreign assets if you expect your own currency to depreciate!
- If you buy 1000 USD when e was 1, but then e goes to 1.1, this means that your 1000 usd now buys 1100 pesos
- So your USD asset will be worth 1100 pesos

Assets-Based Model

- So, you invest in foreign currency **if you expect your own currency to fall later**
- This means, you're buying a currency you expect to stay strong, when you expect your own currency to weaken. Then you can use the stronger currency to buy more of your own currency once it's become weak.
- You will then gain a profit by switching currencies at the right time
- Let's say you buy 1000 dollars for 1000 pesos
- Then, the peso falls to e = 1.1 pesos/dollar
- This means that you can now get 1100 pesos for your 1000 pesos
- You have experienced a 10% capital gain
- They write your expected gain as: (1100-1000)/1000 = 1/10 or 10%
- They write it as $(e^e – e) / e$
- And they then say that your expected return in the US is the US interest rate (base return), plus your capital gain of $(e^e-e)/e$

Finally, your expected total rate of return on dollar-denominated assets is the sum of the interest rate and the expected rate of depreciation of the peso. We denote this *total expected rate of return on dollar-denominated assets* as R^e_{US}. This is given as follows:

$$R^e_{US} = r_{US} + \frac{(e^e - e)}{e} \tag{15.4}$$

As we have just seen, whenever R^e_M and R^e_{US} are not equal, there will be reason for you to reallocate your portfolio between dollar-denominated and peso-denominated assets. These reallocations cause the buying of one currency and the selling of another. *Equilibrium* in the foreign exchange market, in the sense that there is no reason for you (or any other investors) to reallocate your portfolio, requires that $R^e_M = R^e_{US}$. This entails that Equations 15.3 and 15.4 are equal to one another:

$$r_M = r_{US} + \frac{(e^e - e)}{e} \tag{15.5}$$

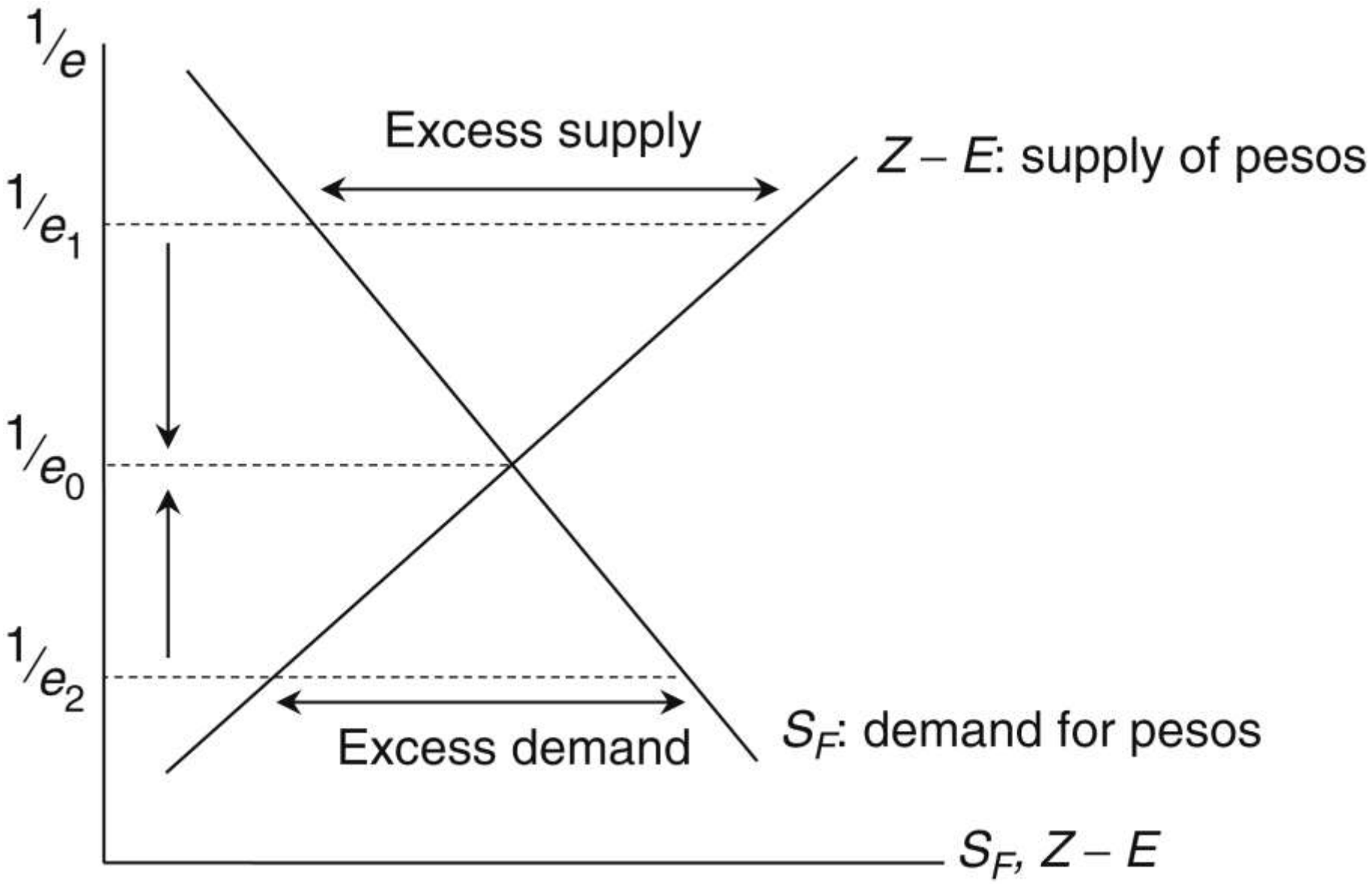
Assets-Based Model

This equation is known as the **interest rate parity condition.**[8] It states that equilibrium in the foreign exchange market requires that the interest rate on peso deposits equals the interest rate on dollar deposits plus the expected rate of peso depreciation. Because it is one of the most important relationships in international finance, we have put it in a box to help you remember[9]:

Interest Rate Parity Condition

$$r_M = r_{US} + \frac{(e^e - e)}{e} \quad \text{Mexico/United States}$$

$$r_H = r_F + \frac{(e^e - e)}{e} \quad \text{Home/Foreign}$$

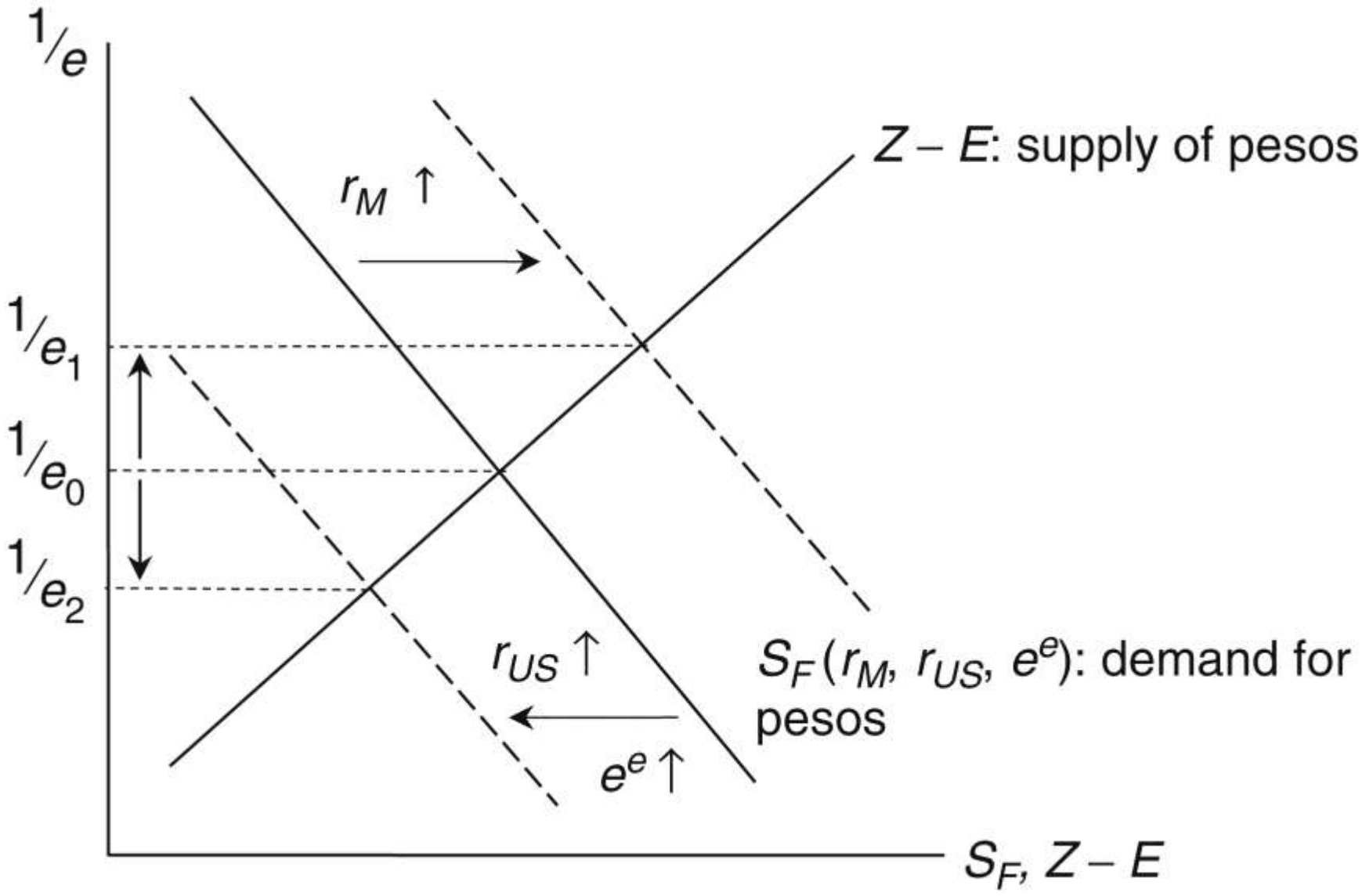

1/e
1/e1
1/e0
1/e2
Excess supply
Z – E: supply of pesos
Excess demand
SF: demand for pesos
SF, Z – E

Putting it all together…

- We now have a more complex and balanced view of how equilibrium is achieved in exchange rate markets
- We have the supply being affected by the trade deficit, and affecting it in turn
- We have the demand being affected by the exchange rate
- As the value goes down, investors demand more (because they expect it to appreciate) (i.e. they expect the foreign currency to become stronger)
- As the value goes up, they demand less (i.e. they expect the foreign currency to become weaker.)
- This is because currencies tend to trade within bands, and so they expect appreciation or depreciation based on its current position within that band
- If it is known to be high, they expect it to be low, and vice versa

The Effect of Interest Rates

- In this model, interest rates are external to our X of Foreign savings/Trade deficit, and our Y of the value of the peso
- So if interest rates change, then we expect this to create a shift in a shift in demand for your currency
- In this model, we are saying that as Mexican interest rates go up, then demand for pesos goes up, because foreign savings will enter the country, demanding to be turned into pesos
- They will buy pesos, and the demand curve will shift to the right
- However, if Mexican rates stay the same, and US interest rates go up, then we expect both Mexicans and foreigners to sell pesos, and shift their money to the US, meaning that we will see a shift in the demand curve for pesos to the left.
- In that case, the peso will depreciate
- In both cases, we're saying that supply is due once more to Z-E (Mexicans sell pesos and buy dollars when they are importing a lot)
- So, if you raise interest rates, then you expect 1/e to rise
- And vice versa


1/e
Z – E: supply of pesos
rM ↑
1/e1
1/e0
1/e2
rUS ↑
SF (rM, rUS, ee): demand for pesos
ee ↑
SF, Z – E

Fixed Exchange Rates

- We now turn to Chapter 18.
- So far, our models have been assuming that all currencies float
- This means, that markets determine their value
- In fact, while most major global currencies do float, it is true that many global currencies still do not
- And/or they are only partially floating
- Until 1971, even the USD and other European currencies were pegged to gold
- Now, however, ‘floating’ is more normative
- Still, major currencies such as the Chinese Yuan are fixed (in this case, pegged to the USD)

Fixed Exchange Rates

- Most small countries, e.g. island countries, simply fix their currencies to a larger currency
- Some larger currencies are also fixed:
- The Danish krone is pegged to the EUR
- As is the Bulgarian lev
- The Jordanian dollar is pegged to the USD
- As is the Saudi riyal
- The Nepalese rupee is pegged to the Indian rupee

Fixed Exchange Rates

- But there is not simply a dichotomy between fixed and floating
- There is in fact a whole spectrum of exchange rate types out there
- This has to do with how much exchange rates are managed by the central government
- Governments can fix exchange rates entirely
- They can adopt a foreign currency as their own
- Or, they can allow for various degrees of ‘managed’ floating

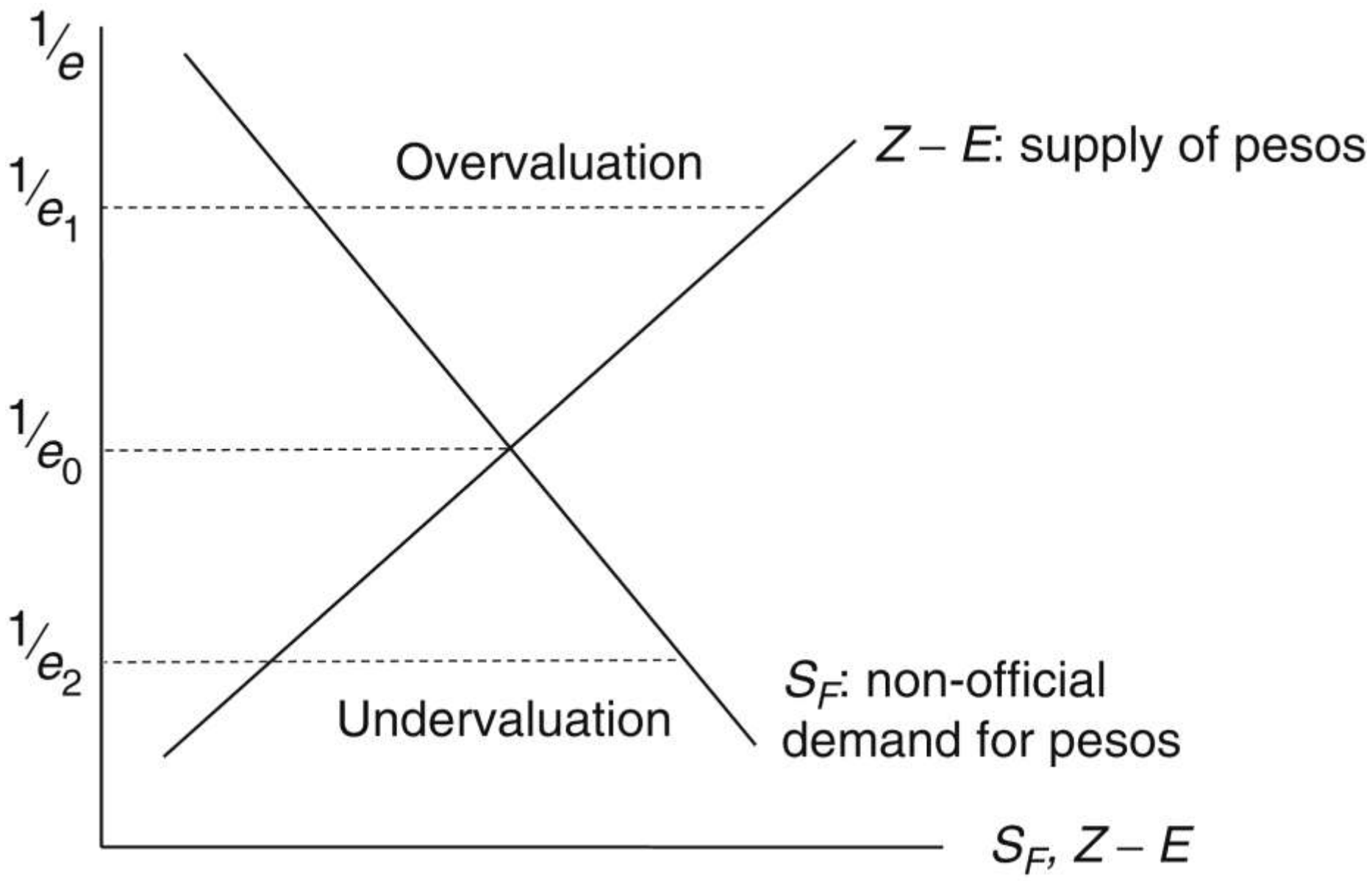
Category	Arrangement	Description	Number of countries
Floating	Flexible (clean) float	The exchange rate is market-determined.	31
Floating	Managed (dirty) float	The exchange rate is primarily market-determined, but the country's monetary authority intervenes in the currency market to influence the movements of the exchange rate.	38
Residual	Other managed arrangements	Volatile currency market conditions prevent the use of any clearly defined exchange rate arrangement.	18
Soft peg	Crawling bands	The country's monetary authority intervenes to maintain the exchange rate in a band around a central rate, and these bands are periodically adjusted.	1
Soft peg	Crawling pegs	The exchange rate is fixed in value to another currency or to a "basket" of other currencies but is adjusted periodically by small amounts.	13
Soft peg	Stabilized arrangement	The exchange rate is officially fixed in value to another currency or to a "basket" of other currencies but actual practice might deviate from this.	24
Hard peg	Fixed or conventional peg	The exchange rate is fixed in value to another currency or to a "basket" of other currencies.	43
Hard peg	Currency board	The exchange rate is fixed in value to another currency, and the domestic currency is fully backed by reserves of this foreign currency.	11
Hard peg	No separate legal tender	The legal tender of the country is a currency of another country.	13

Fixed Exchange Rates

- So how do we model fixed exchange rates?
- One important thing is our terminology
- Whereas in a **float** we talk about **appreciation** and **depreciation**
- In a **fixed** regime, we talk about **devaluation** and **revaluation** (the government devalues, and revalues a currency)
- Governments sometimes like to overvalue their currency, so that they can afford more foreign assets, or so they can sell petrol for more dollars
- Or, they will undervalue their currency, to encourage exports

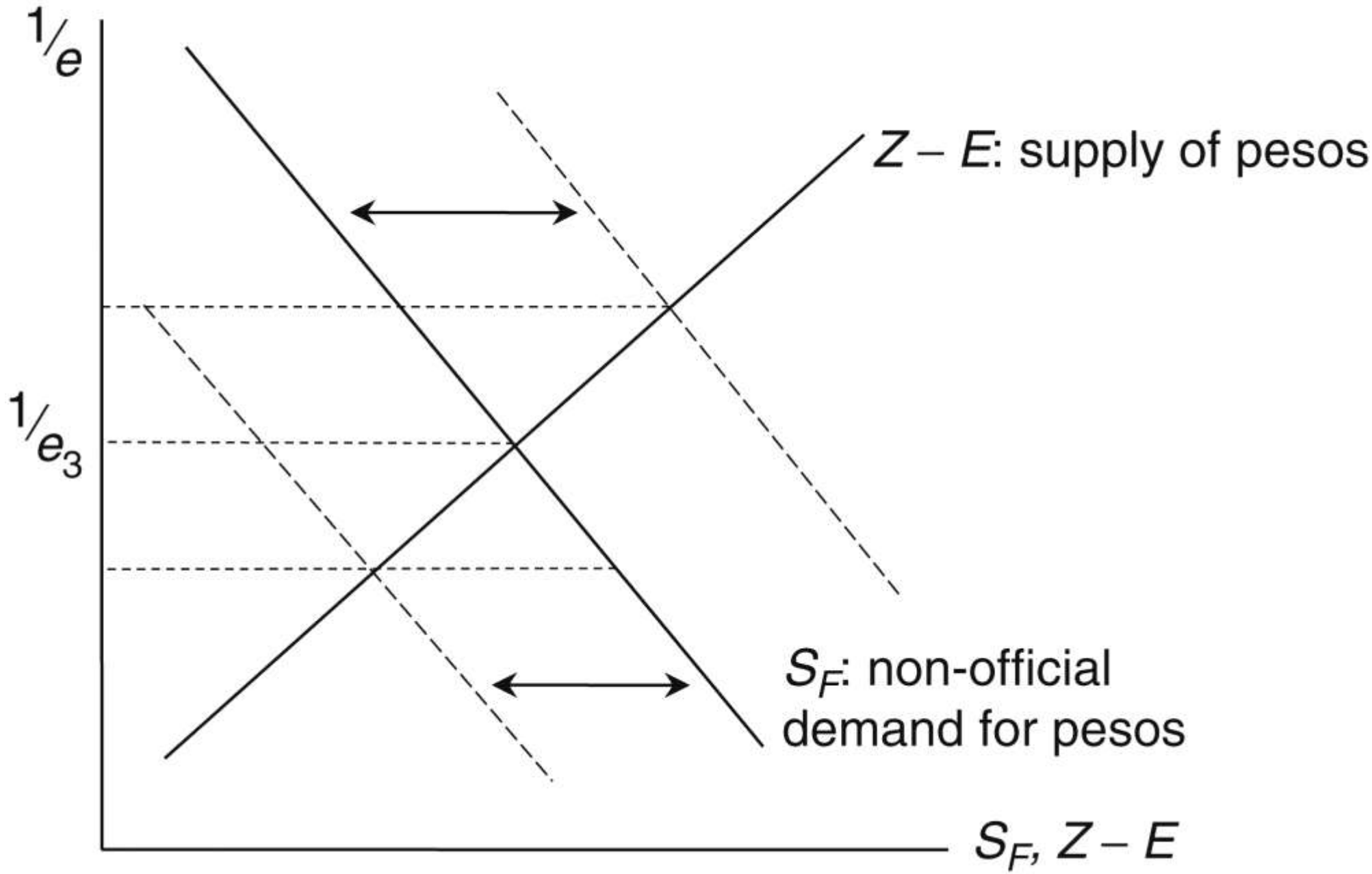
Fixed Exchange Rates

- Whether a government wants it or not, there is always an international market value for a currency
- If a government fixes a currency, it is usually at odds with the market price
- If a government overvalues a currency, this means that there is excess supply relative to demand
- So how does a government maintain the value of a currency at the official overvalued peg?
- It often does so by ‘drawing down foreign reserves’
- You use foreign reserves to buy your own currency, artificially stimulating demand for your currency and taking the excess supply off the market
- So to use our previous model of foreign savings vs. exchange rate:


1/e
1/e1
1/e0
1/e2
Overvaluation
Undervaluation
Z – E: supply of pesos
SF: non-official demand for pesos
SF, Z – E

Interest Rates and Fixed Exchange Rates

- Another major experience that rate fixing countries have is that they have to maintain interest rates equal to what their effective peg has
- Ie., if they have higher interest rates than their peg country, or main trading partner, then investors will flock there and demand their currency, driving up its value
- If they have lower interest rates, then investors will flee that currency and this will drive down its value
- So if a home country wants to maintain an equilibrium fixed exchange rate, it must set its interest rate equal to that prevailing in the foreign country whose currency serves as a peg for the home-country currency
- This is illustrated in the following graph (where higher interest rates in foreign countries, shifts the demand curve left, and vice versa)


1/e
1/e3
Z – E: supply of pesos
SF: non-official
demand for pesos
SF, Z – E

Credibility

- Credibility plays a major role in determining exchange rates
- Investors' expectations are a key player
- If they expect appreciation or depreciation, they will act on it
- Part of expectations is determining if a government will really work to defend a currency
- If e.g. a currency is overvalued, but governments don't want to raise interest rates to keep capital in the country, then investors will flee, expecting a drop
- (Governments might not want to raise interest rates, because this causes business to slow, and unemployment to increase)
- (Read the box: Defending the Brazilian Real on p. 409 for a good illustration of a basic scenario, which happens quite often internationally)

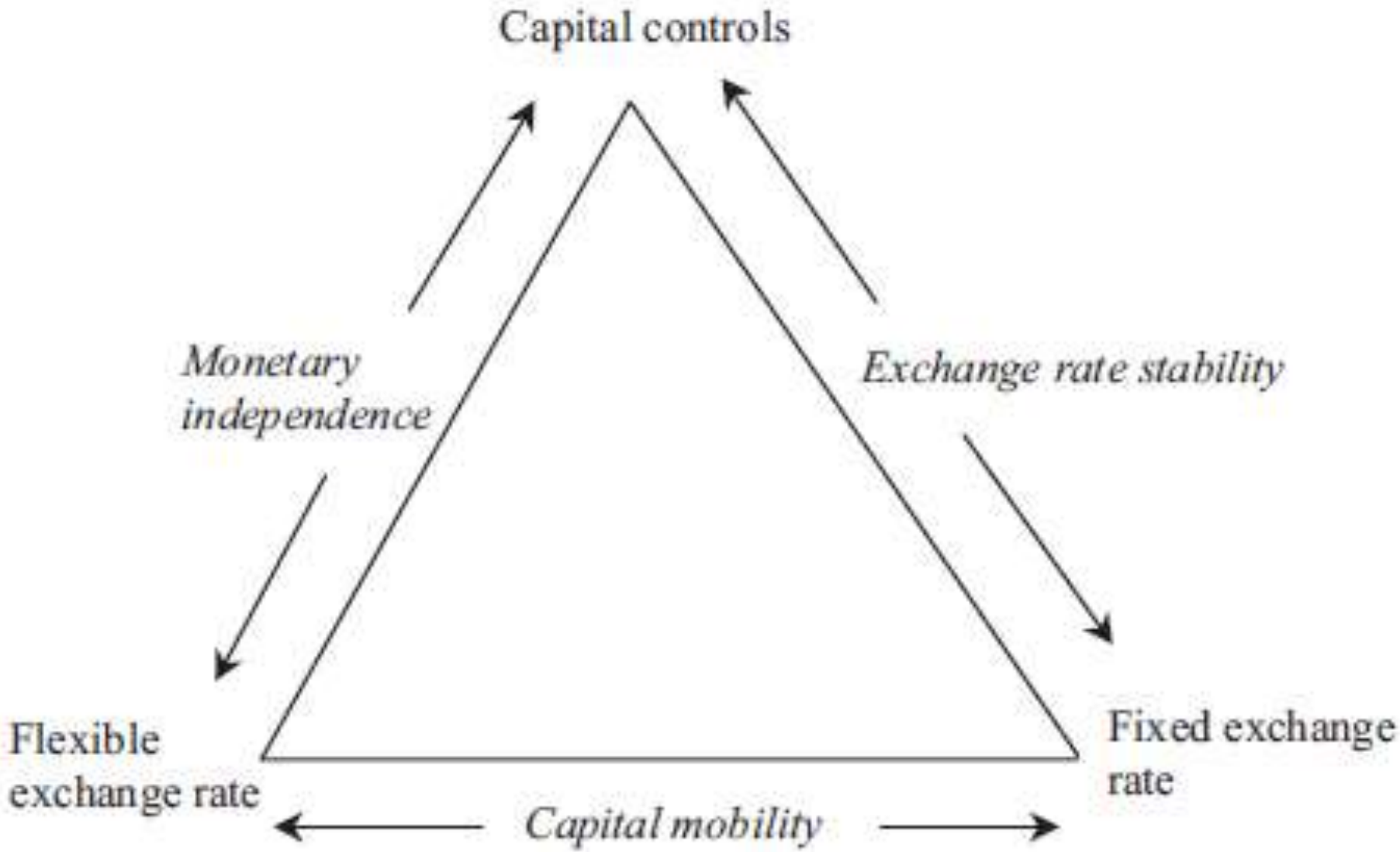


Figure 16.3. The Impossible Trinity

The Impossible Trinity

- This is a theory about monetary policy and exchange rates
- Countries tend to want three things in this realm, but unfortunately, the laws of economics dictate that they can only have 2 out of 3
- We call this situation the Impossible Trinity
- They tend to want:

1. Monetary Independence
2. Exchange rate stability
3. Capital mobility (allow foreign savings/foreign investment)

- Note that larger economies have an easier time of this, but smaller economies have a more difficult time, because they are more affected by foreign capital movements

Impossible Trinity

- In this graph, the point opposite each side of the triangle is the “opposite” thing of what you want to have.
- But the important thing is that any two sides on the triangle, might lead to a problem with the third side of the triangle
- So, monetary independence and exchange rate stability, might lead to a problem if capital is too mobile (mobile capital can cause exchange rate instability)
- And capital mobility, if you want exchange rate stability, might cause a problem with monetary independence, because capital mobility can cause exchange rate instability, so you might have to have fixed exchange rates, or peg your currency to another currency, thus losing monetary independence, if you want both capital mobility and exchange rate stability

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